

Software Development Life Cycle -- Guided Notes

Unit tp-4.1 | CompTIA Network+ N10-009 Obj. FC0-U71 4.1 | N10-008 FC0-U71 4.1

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

The SDLC (Software Development Life Cycle) is a structured process for planning, creating, and maintaining software. The six main phases are: _____, _____, _____, _____, _____, and _____.

Question 2

In the _____ phase, stakeholders define what the software must do. These are captured as functional requirements (what it _____) and non-functional requirements (how well it _____, such as performance and security). Unclear requirements at this stage lead to _____ later.

Question 3

In the _____ phase, developers write code based on the design documents. Code is typically managed in a _____ control system like Git. After coding, the _____ phase verifies the software meets requirements and finds defects before release.

Question 4

The Waterfall model is _____ -- each phase must fully complete before the next begins. It works well when requirements are _____ and unlikely to change. The main risk is that mistakes discovered _____ in the process are very expensive to fix.

Question 5

Agile is an iterative SDLC methodology where work is done in short cycles called _____. At the end of each cycle, a working _____ is delivered and requirements can be _____. Agile prioritizes collaboration with _____ over rigid planning.

Question 6

DevOps combines _____ and _____ teams to automate and integrate the build, test, and deployment pipeline. CI/CD stands for _____ and _____. With CI, every code commit triggers _____ automated tests to catch bugs immediately.

Question 7

Version control (source control) allows developers to track _____ over time. A _____ is a parallel line of development used to add a feature or fix a bug without disturbing the main code. After testing, the branch is _____ back into the main codebase. Git is the most widely used _____ version control system.

Question 8 -- Real World Application

REAL WORLD: A startup uses Waterfall to build their mobile app over 18 months. At the end of development, they discover that users want a completely different onboarding flow than what was designed at the start. All the code must be rewritten. What SDLC methodology would have prevented this and why?

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/tp-4-1-software-development-life-cycle/tp-4-1-software-development-life-cycle-answer-key.pdf
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